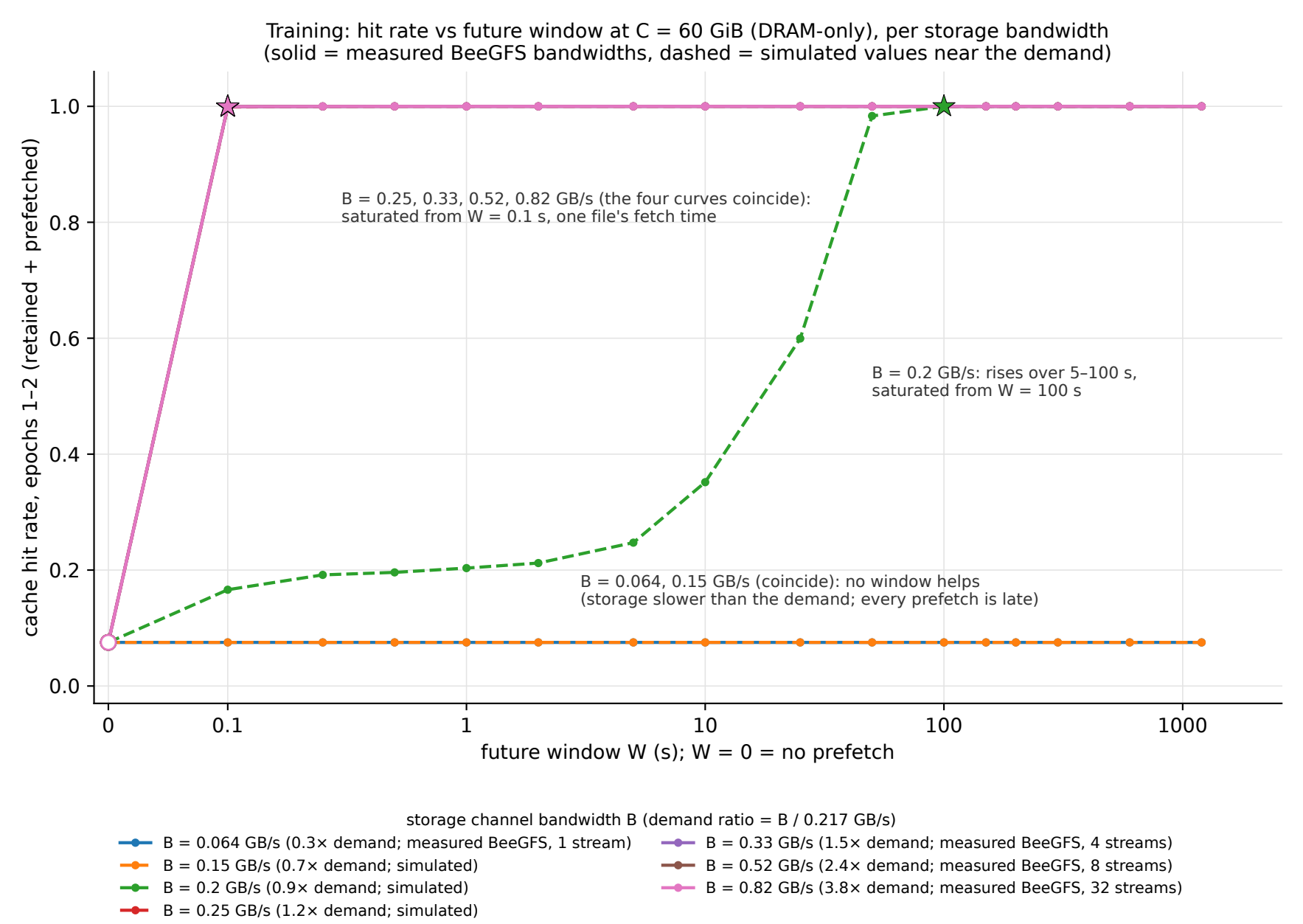

Demand timeline = the REAL CosmoFlow run (A100 node c33, 3 epochs × 61,440 TFRecord files = 178 GB; epochs 1–2 at 74.9 samples/s = 0.217 GB/s). Cache and storage = trace-driven SIMULATION (train_prefetch_sim.py: one FIFO storage channel of bandwidth B, LRU cache with window protection, every epoch's order known at job start; the prefetcher fetches the samples due within the next W seconds). Hit rate = share of the 122,880 sample demands of epochs 1–2 (epoch 0 = cold start) served from the cache: RETAINED from the previous epoch, or PREFETCHED and complete before the consuming training step (a prefetch still in flight counts as a miss). Measured BeeGFS cold-read bandwidths from c33: 0.064 / 0.33 / 0.52 / 0.82 GB/s at 1 / 4 / 8 / 32 streams; 0.15 / 0.20 / 0.25 GB/s are simulated values. After the 7.5% retention of a 60 GiB cache the storage must supply 0.20 GB/s.